J1A
DIN96 / male right-angle

ROW A

A1	
A2	
A3	GND
A4	+12V_BP
A5	+12V_BP
A6	GND
A7	+5V_BP
A8	+5V_BP
A9	GND
A10	
A11	
A12	
A13	
A14	
A15	
A16	HOST_SCL
A17	
A18	
A19	
A20	HOST_RX
A21	
A22	
A23	
A24	GND
A25	+5V_BP
A26	+5V_BP
A27	GND
A28	+12V_BP
A29	+12V_BP
A30	GND
A31	
A32	

J1B
DIN96 / male right-angle

ROW B

B1	GND
B2	GND
B3	GND
B4	+12V_BP
B5	+12V_BP
B6	GND
B7	+5V_BP
B8	+5V_BP
B9	GND
B10	GND
B11	GND
B12	GND
B13	GND
B14	GND
B15	GND
B16	GND
B17	GND
B18	GND
B19	GND
B20	GND
B21	GND
B22	GND
B23	GND
B24	GND
B25	+5V_BP
B26	+5V_BP
B27	GND
B28	+12V_BP
B29	+12V_BP
B30	GND
B31	GND
B32	GND

J1C
DIN96 / male right-angle

ROW C

C1	
C2	
C3	GND
C4	+12V_BP
C5	+12V_BP
C6	GND
C7	+5V_BP
C8	+5V_BP
C9	GND
C10	
C11	
C12	
C13	
C14	
C15	
C16	HOST_SDA
C17	
C18	
C19	
C20	HOST_TX
C21	
C22	
C23	
C24	GND
C25	+5V_BP
C26	+5V_BP
C27	GND
C28	+12V_BP
C29	+12V_BP
C30	GND
C31	
C32	

Existing backplane interface: 12 V, 5 V, UART and I2C0.

No new pin assignments. All 24 V contacts remain unused.

USBInterface

USB_VBUS	USB_VBUS
USB_D+	USB_D+
USB_D−	USB_D−
USB_CC1	USB_CC1
USB_CC2	USB_CC2
USB_VBUS_SENSE	USB_VBUS_SENSE

File: USBInterface.kicad_sch

Power

+5V_BP	+5V_BP
+12V_BP	+12V_BP
+3V3	+3V3
+5V_LOAD	+5V_LOAD
+12V_MOTOR	+12V_MOTOR
POWER_12V_SENSE	POWER_12V_SENSE

File: Power.kicad_sch

MotionController

+3V3	+3V3
RP_RUN	RP_RUN
RP_SWCLK	RP_SWCLK
RP_SWDIO	RP_SWDIO
USB_D+	USB_D+
USB_D−	USB_D−

File: MotionController.kicad_sch

MotionGPIO

+3V3	+3V3
ENC1_CS	ENC1_CS
ENC2_CS	ENC2_CS
ENC3_CS	ENC3_CS
ENC_MISO	ENC_MISO
ENC_MOSI	ENC_MOSI
ENC_SCK	ENC_SCK
HEATER_PWM	HEATER_PWM
HOST_RX	HOST_RX
HOST_SCL	HOST_SCL
HOST_SDA	HOST_SDA
HOST_TX	HOST_TX
M1_A1	M1_A1
M1_A2	M1_A2
M1_B1	M1_B1
M1_B2	M1_B2
M2_A1	M2_A1
M2_A2	M2_A2
M2_B1	M2_B1
M2_B2	M2_B2
M3_A1	M3_A1
M3_A2	M3_A2
M3_B1	M3_B1
M3_B2	M3_B2
MOTOR_EN	MOTOR_EN
MOTOR_PWM	MOTOR_PWM
POWER_12V_SENSE	POWER_12V_SENSE
ROT_A1	ROT_A1
ROT_A2	ROT_A2
ROT_B1	ROT_B1
ROT_B2	ROT_B2
ROT_CS	ROT_CS
ROT_ENABLE	ROT_ENABLE
ROT_ENDSTOP	ROT_ENDSTOP
ROT_MISO	ROT_MISO
ROT_MOSI	ROT_MOSI
ROT_SCK	ROT_SCK
TEMP_IN	TEMP_IN
TOOL_PWM1	TOOL_PWM1
TOOL_PWM2	TOOL_PWM2
USB_VBUS_SENSE	USB_VBUS_SENSE
VAC_PWM	VAC_PWM

File: MotionGPIO.kicad_sch

MotionDrivers

+3V3	+3V3
+12V_MOTOR	+12V_MOTOR
M1_A1	M1_A1
M1_A2	M1_A2
M1_AO1	M1_AO1
M1_AO2	M1_AO2
M1_B1	M1_B1
M1_B2	M1_B2
M1_BO1	M1_BO1
M1_BO2	M1_BO2
M2_A1	M2_A1
M2_A2	M2_A2
M2_AO1	M2_AO1
M2_AO2	M2_AO2
M2_B1	M2_B1
M2_B2	M2_B2
M2_BO1	M2_BO1
M2_BO2	M2_BO2
M3_A1	M3_A1
M3_A2	M3_A2
M3_AO1	M3_AO1
M3_AO2	M3_AO2
M3_B1	M3_B1
M3_B2	M3_B2
M3_BO1	M3_BO1
M3_BO2	M3_BO2
MOTOR_EN	MOTOR_EN
MOTOR_PWM	MOTOR_PWM

File: MotionDrivers.kicad_sch

J13
ENDSTOP

1	ROT_ENDSTOP
2	+3V3
3	GND

J3
MOTOR 1

1	M1_BO1
2	M1_BO2
3	M1_AO2
4	M1_AO1

J4
MOTOR 2

1	M2_BO1
2	M2_BO2
3	M2_AO2
4	M2_AO1

J5
MOTOR 3

1	M3_BO1
2	M3_BO2
3	M3_AO2
4	M3_AO1

J6
MOTION ENCODERS

1	GND
2	+3V3
3	ENC_MOSI
4	ENC_MISO
5	ENC_SCK
6	ENC1_CS
7	ENC2_CS
8	ENC3_CS

J12
ROTATION ENCODER

1	+3V3
2	ROT_MISO
3	ROT_MOSI
4	ROT_SCK
5	ROT_CS
6	GND

J14
ROTATION MOTOR

1	ROT_OUT1
2	ROT_OUT2
3	ROT_OUT3
4	ROT_OUT4

J19
TOOL PWM

1	GND
2	TOOL_PWM1
3	TOOL_PWM2
4	+3V3

J15
HEATER + TEMP

1	HEATER_5V
2	HEAT_LOW
3	+3V3
4	GND
5	TEMP_IN

J16
VACUUM / 5 V

1	+5V_LOAD
2	VAC_LOW

J17
THERMAL CUTOFF

1	+5V_LOAD
2	HEATER_5V

J18
USB-C / SERVICE

VBUS	A4 USB_VBUS
CC1	A5 USB_CC1
CC2	B5 USB_CC2
D−	A7 USB_D−
D−	B7
D+	A6 USB_D+
D+	B6
SBU1	A8
SBU2	B8
SHIELD	SH
GND	A1

GND

J2
SWD / RP2350

1	+3V3
2	RP_SWDIO
3	GND
4	RP_SWCLK
5	RP_RUN

ToolOutputs

+5V_BP	+5V_BP
+3V3	+3V3
+5V_LOAD	+5V_LOAD
HEATER_PWM	HEATER_PWM
HEAT_LOW	HEAT_LOW
ROT_A1	ROT_A1
ROT_A2	ROT_A2
ROT_B1	ROT_B1
ROT_B2	ROT_B2
ROT_ENABLE	ROT_ENABLE
ROT_OUT1	ROT_OUT1
ROT_OUT2	ROT_OUT2
ROT_OUT3	ROT_OUT3
ROT_OUT4	ROT_OUT4
VAC_LOW	VAC_LOW
VAC_PWM	VAC_PWM

File: ToolOutputs.kicad_sch

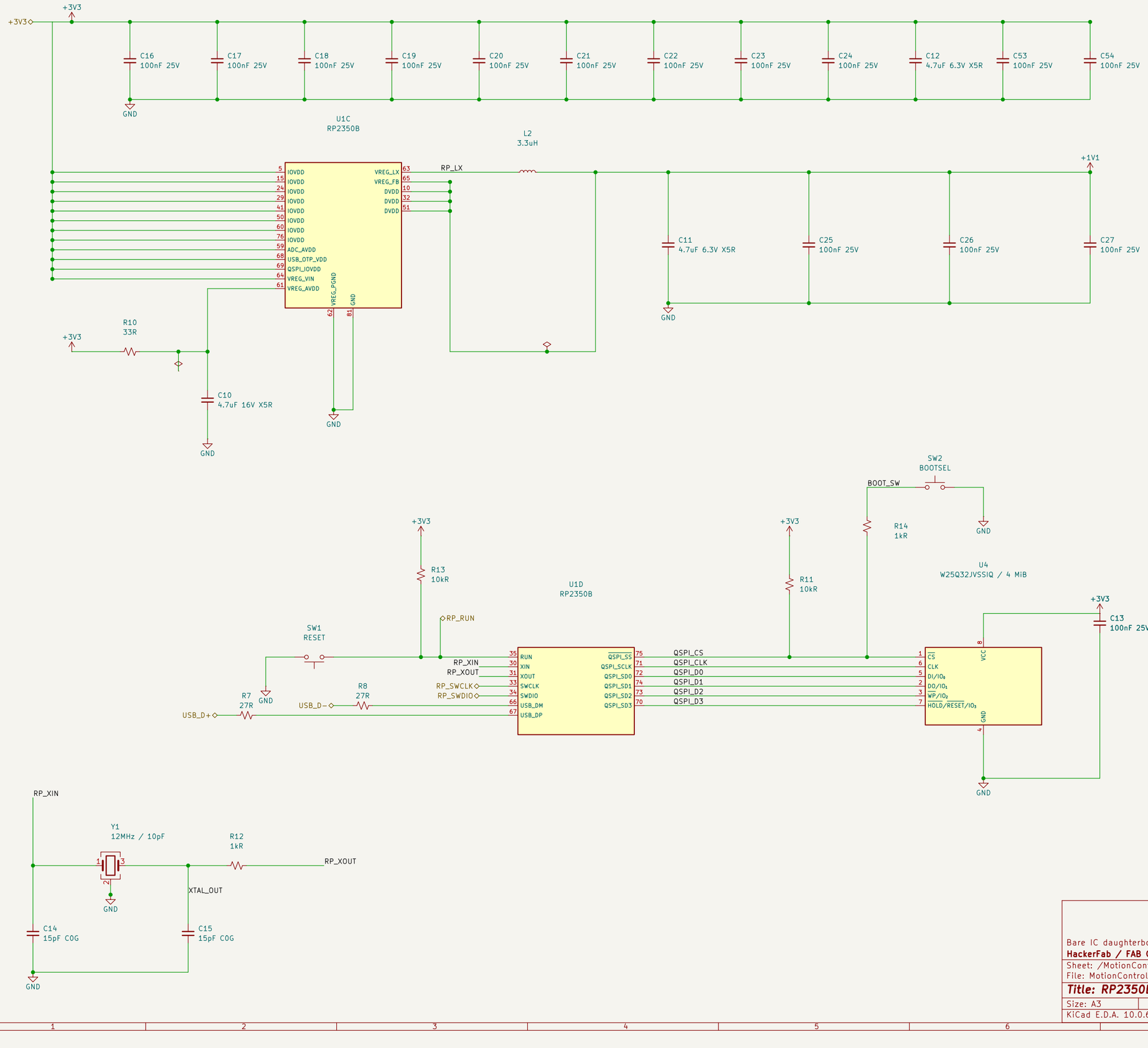
USB: data only; power the backplane before service.

Bare IC daughterboard / engineering review
HackerFab / FAB Computer

Sheet: /
File: FabNanoDaughterboard.kicad_sch

Title: FAB Nano

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/7



Bare IC daughterboard / engineering review

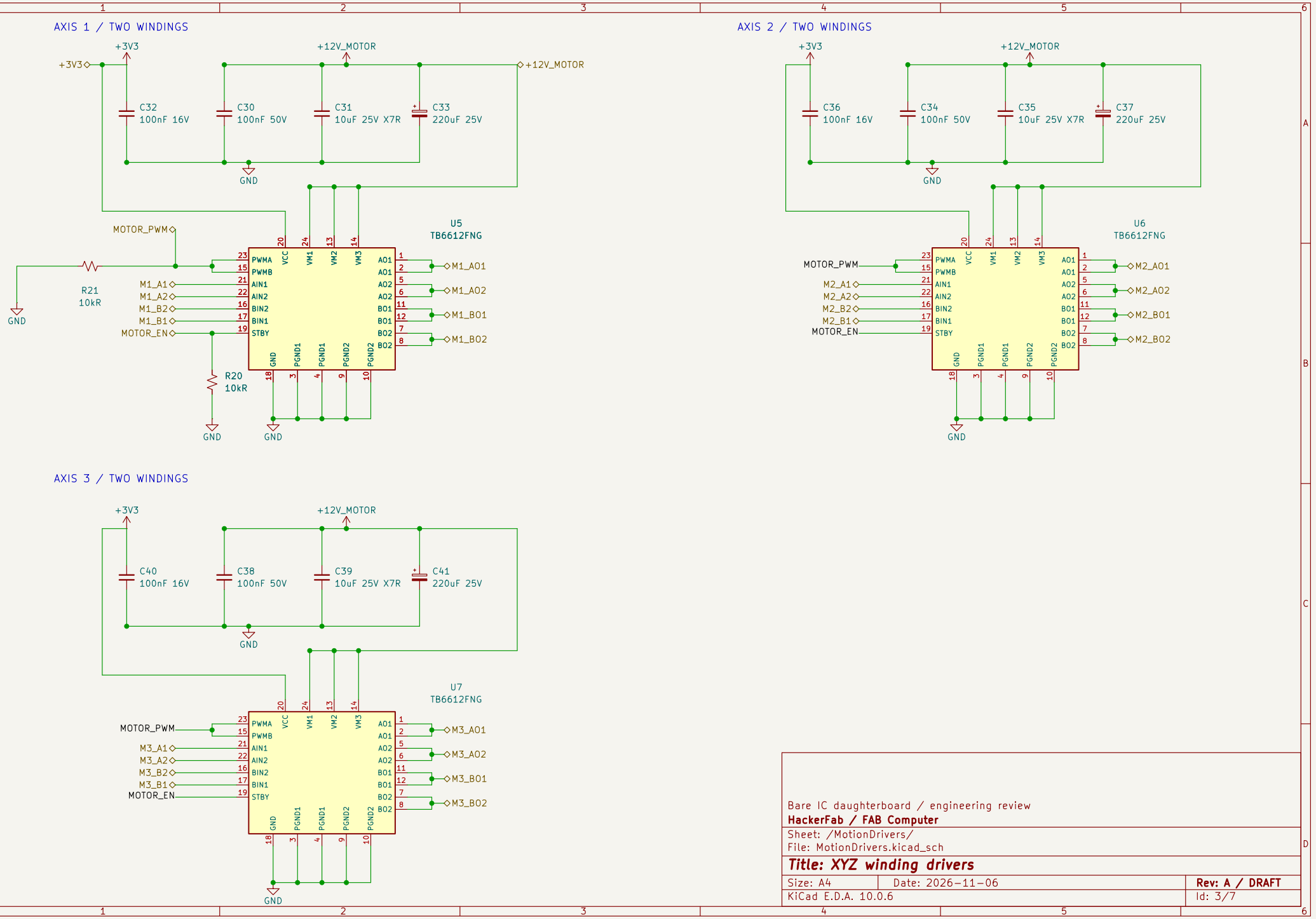
HackerFab / FAB Computer

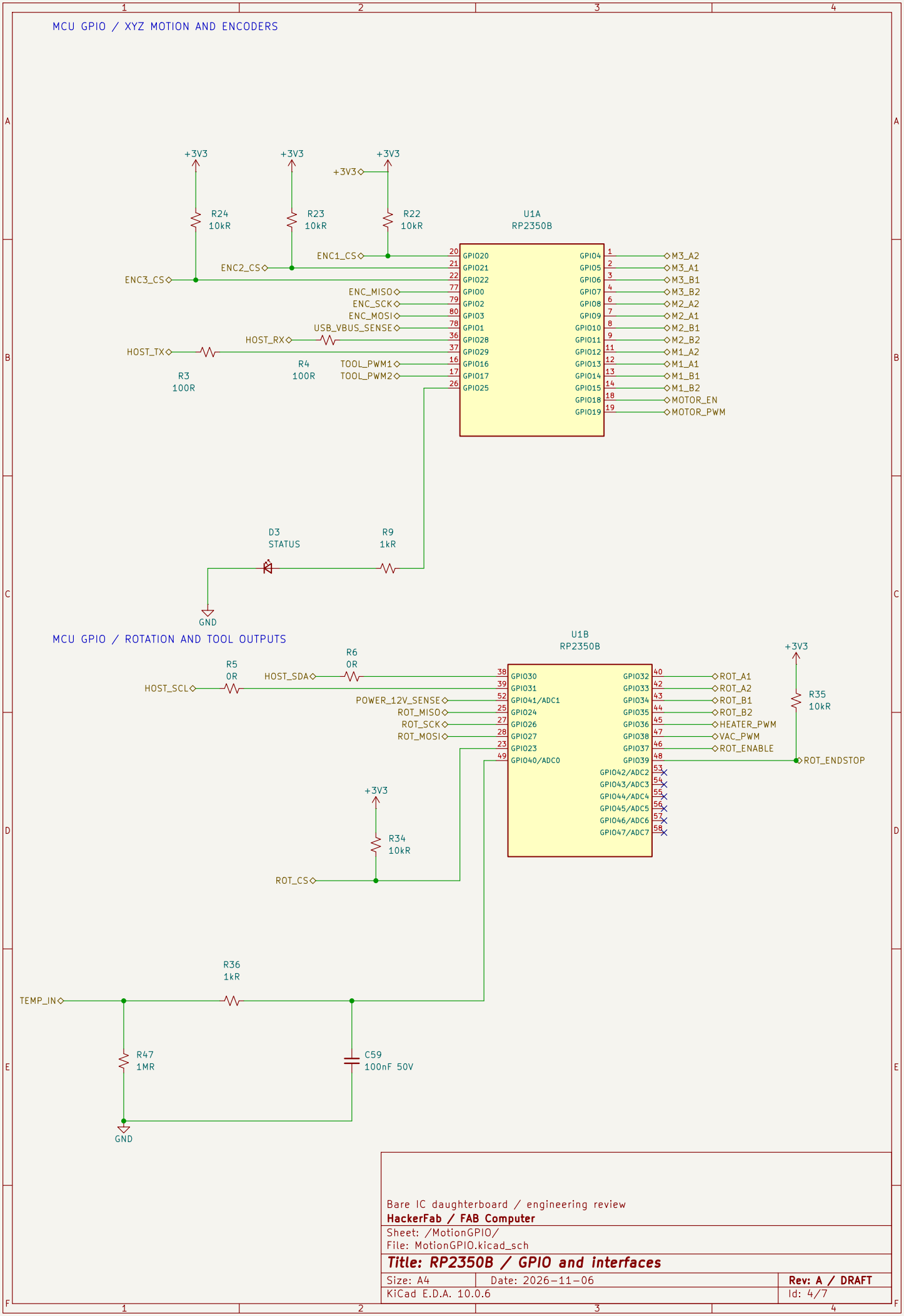
Sheet: /MotionController/

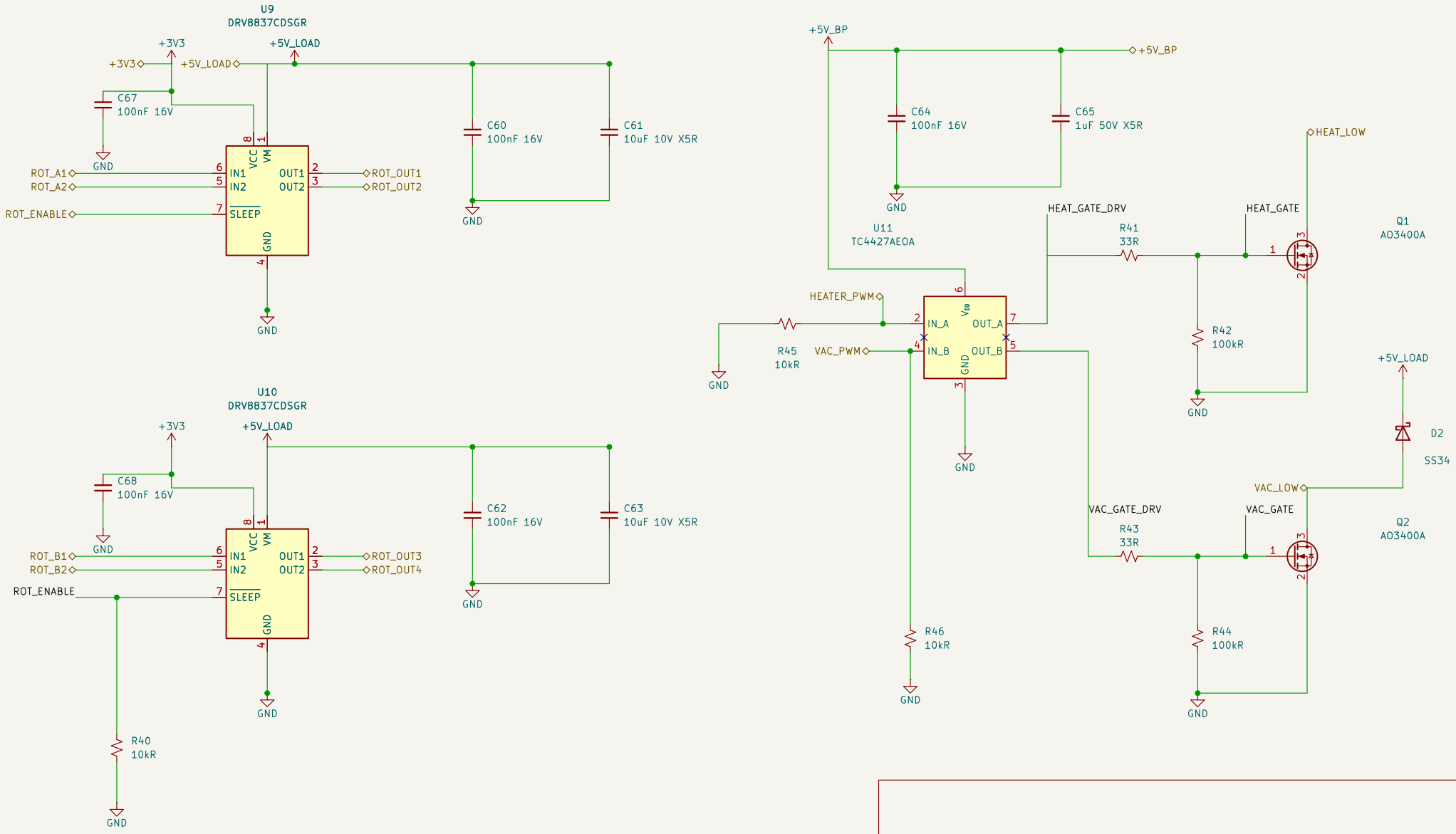
File: MotionController.kicad_sch

Title: RP2350B

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 2/7


AXIS 1 / TWO WINDINGS
+3V3
+12V_MOTOR
C32
100nF 16V
C30
100nF 50V
C31
10uF 25V X7R
C33
220uF 25V
GND
U5
TB6612FNG
MOTOR_PWM
R21
10kR
M1_A1
M1_A2
M1_B2
M1_B1
MOTOR_EN
R20
10kR
M1_AO1
M1_AO2
M1_BO1
M1_BO2
AXIS 2 / TWO WINDINGS
C36
100nF 16V
C34
100nF 50V
C35
10uF 25V X7R
C37
220uF 25V
U6
TB6612FNG
M2_A1
M2_A2
M2_B2
M2_B1
M2_AO1
M2_AO2
M2_BO1
M2_BO2
AXIS 3 / TWO WINDINGS
C40
100nF 16V
C38
100nF 50V
C39
10uF 25V X7R
C41
220uF 25V
U7
TB6612FNG
M3_A1
M3_A2
M3_B2
M3_B1
M3_AO1
M3_AO2
M3_BO1
M3_BO2
Bare IC daughterboard / engineering review
HackerFab / FAB Computer
Sheet: /MotionDrivers/
File: MotionDrivers.kicad_sch
Title: XYZ winding drivers
Size: A4
Date: 2026-11-06
Rev: A / DRAFT
KiCad E.D.A. 10.0.6
Id: 3/7


MCU GPIO / XYZ MOTION AND ENCODERS
+3V3
+3V3
+3V3
+3V3
R24
10kR
R23
10kR
R22
10kR
U1A
RP2350B
ENC1_CS
ENC2_CS
ENC3_CS
ENC_MISO
ENC_SCK
ENC_MOSI
USB_VBUS_SENSE
HOST_RX
HOST_TX
TOOL_PWM1
TOOL_PWM2
R3
100R
R4
100R
GPIO20
GPIO21
GPIO22
GPIO0
GPIO2
GPIO3
GPIO1
GPIO28
GPIO29
GPIO16
GPIO17
GPIO25
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO18
GPIO19
M3_A2
M3_A1
M3_B1
M3_B2
M2_A2
M2_A1
M2_B1
M2_B2
M1_A2
M1_A1
M1_B1
M1_B2
MOTOR_EN
MOTOR_PWM
D3
STATUS
R9
1kR
GND
MCU GPIO / ROTATION AND TOOL OUTPUTS
U1B
RP2350B
R5
0R
R6
0R
HOST_SCL
HOST_SDA
POWER_12V_SENSE
ROT_MISO
ROT_SCK
ROT_MOSI
GPIO30
GPIO31
GPIO41/ADC1
GPIO24
GPIO26
GPIO27
GPIO23
GPIO40/ADC0
GPIO32
GPIO33
GPIO34
GPIO35
GPIO36
GPIO38
GPIO37
GPIO39
GPIO42/ADC2
GPIO43/ADC3
GPIO44/ADC4
GPIO45/ADC5
GPIO46/ADC6
GPIO47/ADC7
ROT_A1
ROT_A2
ROT_B1
ROT_B2
HEATER_PWM
VAC_PWM
ROT_ENABLE
ROT_ENDSTOP
+3V3
R35
10kR
+3V3
R34
10kR
ROT_CS
R36
1kR
TEMP_IN
R47
1MR
C59
100nF 50V
GND
Bare IC daughterboard / engineering review
HackerFab / FAB Computer
Sheet: /MotionGPIO/
File: MotionGPIO.kicad_sch
Title: RP2350B / GPIO and interfaces
Size: A4
Date: 2026-11-06
Rev: A / DRAFT
KiCad E.D.A. 10.0.6
Id: 4/7



Thermal cutoff J17 is in series with heater supply.

Headers and load wiring: top-level connector sheet.

Bare IC daughterboard / engineering review

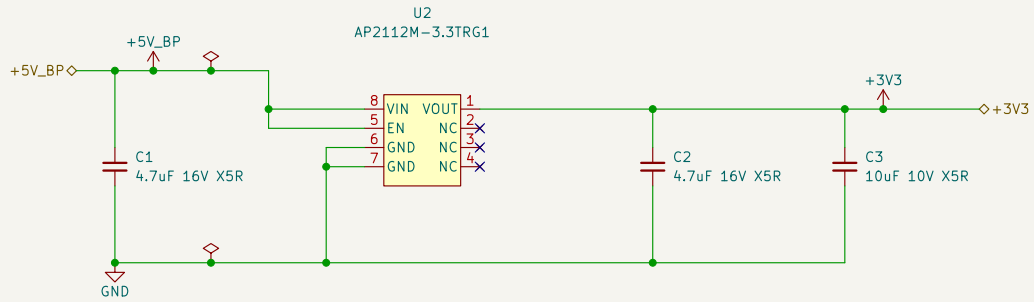
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Sheet: /ToolOutputs/

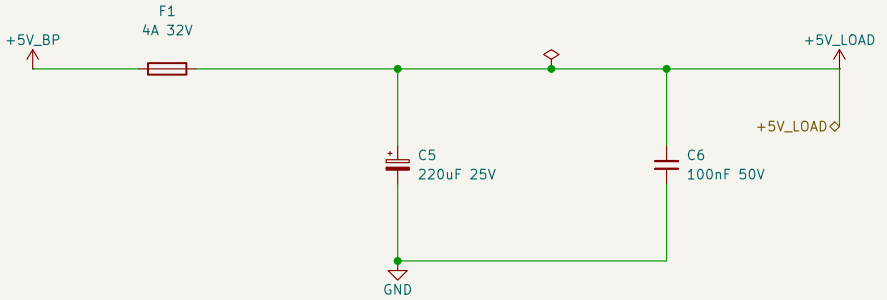
File: ToolOutputs.kicad_sch

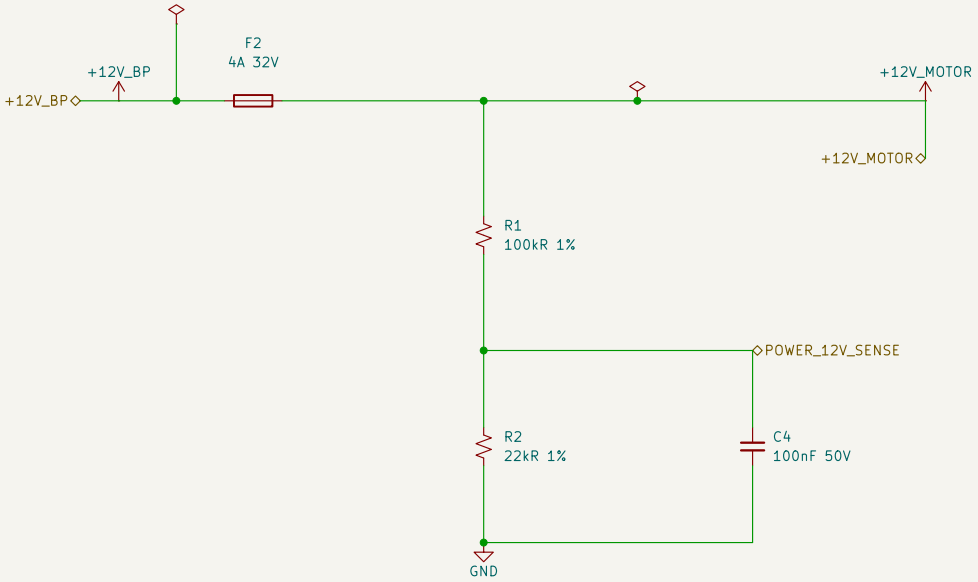
Title: Rotation, heater and vacuum

Size: A4	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 5/7



Logic only. No new backplane contact.





12 V absent: inhibit tool outputs.

Bare IC daughterboard / engineering review

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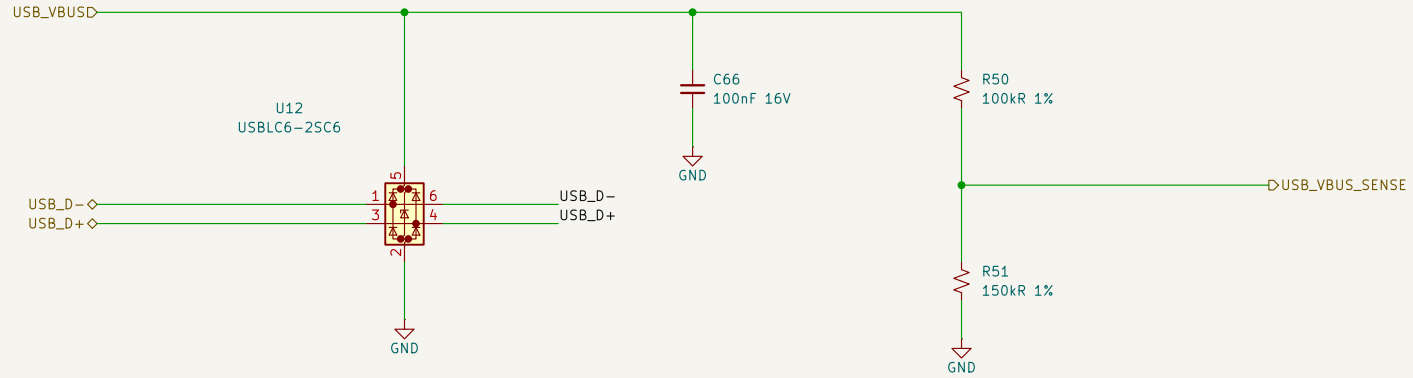
Sheet: /Power/

File: Power.kicad_sch

Title: Backplane rails and interfaces

Size: A4	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 6/7

USB SERVICE / ESD PROTECTION AND VBUS DETECTION



USB-C SINK / CC TERMINATIONS



Self-powered USB 2.0 service port. Backplane power must be present.

CC1 and CC2 have independent 5.1k pull-downs. VBUS feeds protection and sensing only.

Bare IC daughterboard / engineering review

HackerFab / FAB Computer

Sheet: /USBInterface/

File: USBInterface.kicad_sch

Title: USB service interface

Size: A4	Date: 2026-11-07	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 7/7